

Risk Measurement and Financial Institutions

Assignment 1 - Individual

1. Congratulations: on 22nd April 2024, you were appointed to *RMFI Funds Management* as a government bond dealer, with **HUF 100 million** of funds available for buying bonds! The same day (i.e. 22nd April 2024), you will invest this amount (ie HUF 100 million) to buy the bonds and build up your portfolio. Please specify which **Hungarian government securities (at least 3 different types)** you would buy on this day, if the strategy of the bond funds you manage is to have a portfolio with a duration of at least 4 years and at most 7 years! You can use the website www.akk.hu (for basic data <https://www.akk.hu/statisztika/allampapirok/alapadatok>, for historical data <https://www.akk.hu/statisztika/historikus-adatok>). For the portfolio, please provide at least the name, ISIN code, number of units, market value and duration of the bond. (2 points)
 2. **What is the VaR value** of the former portfolio if you want to calculate with exactly the same parameters (holding period, confidence level) as the trading book regulation requires for the internal model method (99%, 10 days)? Use a time series of at least 1 year and the delta-normal method (but all other choices (which yield curve, which mapping method) are up to you). (4 points)
 3. **What would be the value of the portfolio, if yields would be the same as they were on 17th October 2022?** What would have been the % change in value? **Compare it with the VaR** and evaluate the result. (2 points)
 4. The risk management system indicates that the VaR value just calculated exceeds the VaR limit by 20%! Which transaction(s) and in what amount do you need to conclude in order to **bring the VaR below the limit?** (2 points)
- +1. EXTRA question, optional, for EXTRA 2 points: How much of this VaR is attributable to market risk and how much to credit risk? (+ 2 points)

Deadline: 18 May 2024 (Saturday) till 23:59 (11:59pm) (DEADLINE EXTENDED!)

This exercise has to be done **individually, NOT in groups!**

Where to send: by email, to Enikő Lájér, to ELajer@bankarkepzo.hu AND upload to moodle - please do it in both ways! Thank you in advance!

How to submit: in a well-edited, easy to read and understand **MS Excel file** with the filename **[YourName]_RMFIAssignment1.xls**. The same should be included in the 'Subject' field of the email.